

FIT5196 DATA WRANGLING

Week 4

Exploratory Data Analysis

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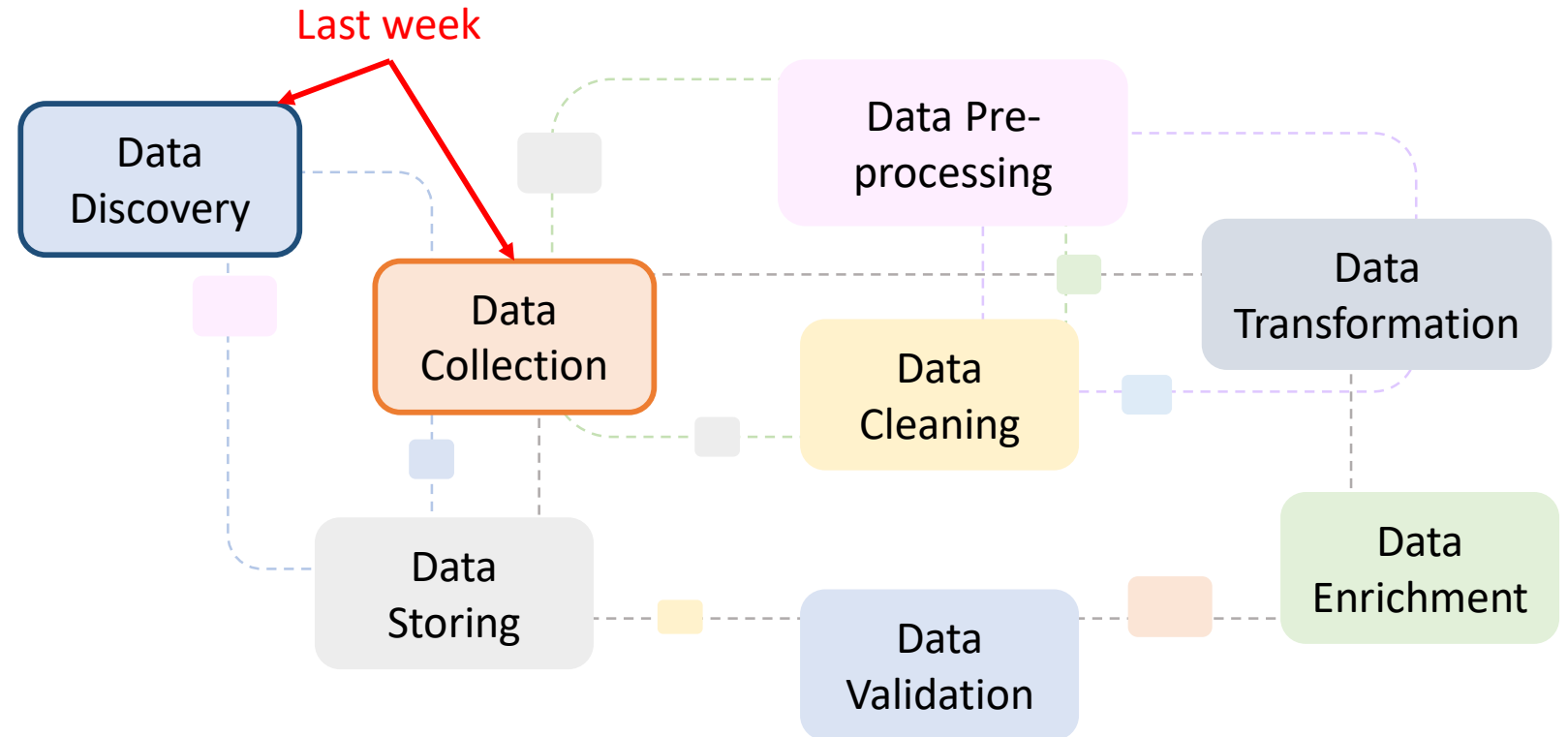
Monash University

Data Wrangling Tasks (Recap)

- **Data Wrangling** is the process of **acquiring**, **cleaning**, **structuring**, and **enriching** raw data into a format that is directly usable for analysis.

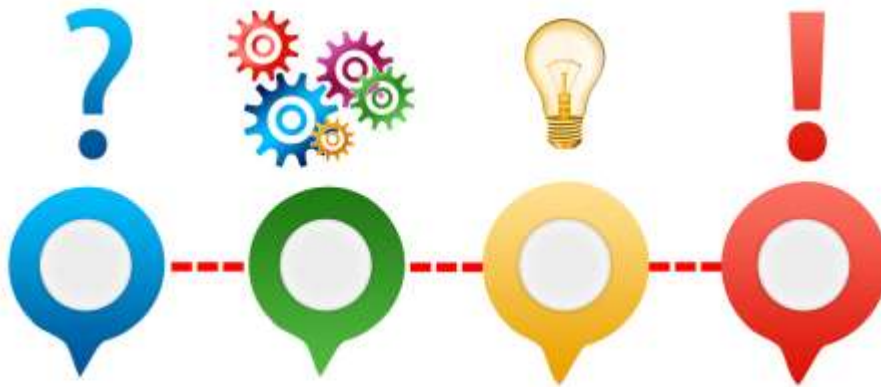
Data discovery is the process of **identifying** and **understanding data sources** that can be used for analytical purposes.

Data collection is the systematic process of **gathering** and **measuring** information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes.



Outline

- Exploratory Data Analysis (EDA)
 - EDA for Numerical Data
 - EDA for Text Data
- Text Data Pre-processing



Exploratory Data Analysis (EDA)

- **Exploratory Data Analysis (EDA)** is a crucial step in the data analysis process, enabling data scientists and analysts to **understand** the **patterns**, **anomalies**, and **relationships** in their data.
- The **goal** of EDA is to **use statistical graphics, plots, and information tables** to explore and summarize the **main characteristics** of the data, often visually, without making any assumptions about its contents.
- This approach helps in identifying the underlying structure, detecting outliers or anomalies, testing assumptions, and developing initial ideas for models.



Common Data Types

- Numerical Data
- Categorical Data
- Text Data
- Image Data



Common Data Types

- **Numerical Data**

- Numerical data represents **quantifiable information** that can be **measured** and **expressed** using **numbers**.

Any value within a range
and can be divided

infinitely

For example,

- Height
- Weight
- Temperature
- Speed
- ...

**Continuous
Data**

Countable values or
categories

For example,

- Student attendance
- Available parking bays
- Number of coding lines
- Size of a data set
- ...

**Discrete
Data**

Common Data Types (cont.)

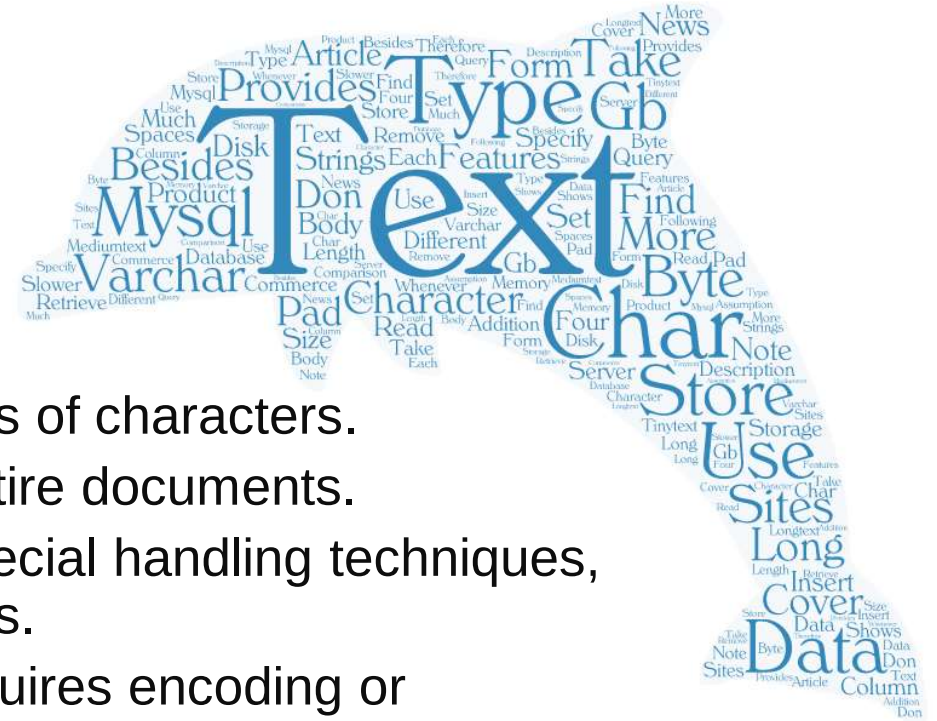
- Numerical Data
- Character Data
 - Often referred to as "char" in programming languages, consists of single **alphabetic letters**, **symbols**, or **numerals** that do not have numeric value.
 - It's the **basic unit** used to construct strings (text) and is typically stored in a fixed-length format.
 - For example,
 - 'A', 'b', '1', '6', or '?', '!', '#'...

ASCII TABLE

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	\$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[END OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Common Data Types (cont.)

- Numerical Data
- Character Data
- **Text Data**
 - Text data is composed of strings, which are sequences of characters.
 - Text can include words, sentences, paragraphs, or entire documents.
 - In data analysis and processing, text data requires special handling techniques, especially for natural language processing (NLP) tasks.
 - Text data is used for qualitative analysis and often requires encoding or tokenization to be effectively used in computational processes.



Source: <https://www.mysqltutorial.org/wp-content/uploads/2017/02/MySQL-TEXT-data-type.png>

Common Data Types (cont.)

- Numerical Data
- Character Data
- Text Data
- **Image Data**
 - Image data represents **visual information** encoded digitally as a bitmap or a set of pixel values.
 - Image data is used in a wide range of applications, from simple photo editing to complex tasks like image recognition and computer vision.
 - Processing and analysing image data often involve techniques from the field of deep learning, particularly convolutional neural networks (CNNs).



Image processing applied on the “Lena” image.

Source: 1972 issue of Playboy Magazine

Understanding Data Types

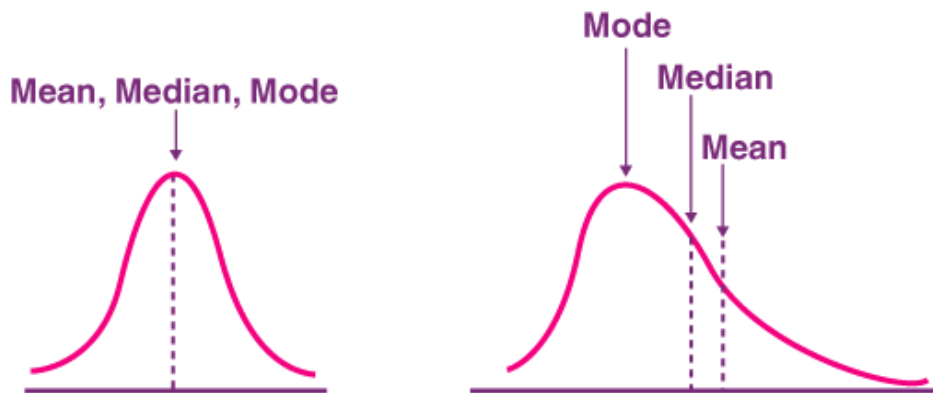
- Identifying data types during Exploratory Data Analysis (EDA) is a fundamental step that significantly influences how data will be handled, processed, and analysed.
- Data types dictate the kind of statistical analysis that can be performed, the visualizations that are appropriate, and the data pre-processing steps needed.
- It is important to understand data types in EDA
 - Appropriate Analysis
 - Effective Visualisation
 - Correct methods for data cleaning and pre-processing
 - Effective feature engineering
 - Suitable algorithms and models for better performance
 - Ensuring data integrity



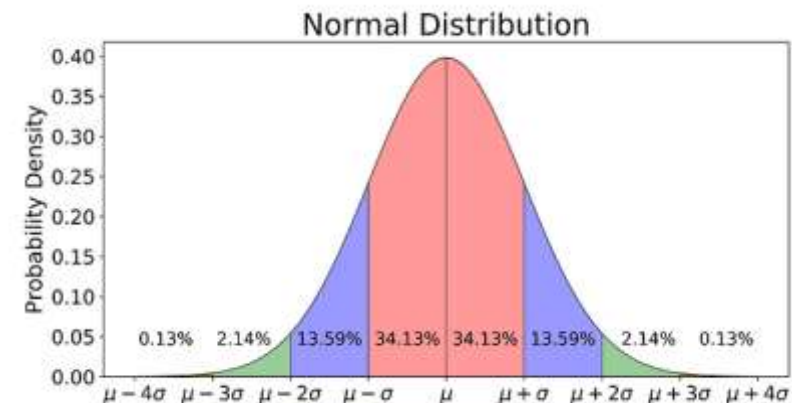
Understanding Data Types (cont.)

- **Appropriate Analysis**

- Different types of data require different statistical methods and tools for analysis.
 - **Numerical data** allows for calculations of mean, median, standard deviation, and other statistical measures.
 - **Categorical data** may be analysed using frequency counts, mode, or chi-square tests.
- Understanding the data type ensures that the analysis conducted is appropriate and meaningful.



Source: <https://cdn1.byjus.com/wp-content/uploads/2022/09/Measures-Of-Central-Tendency-Mean-Median-and-Mode.png>



Source: https://miro.medium.com/v2/resize:fit:720/format:webp/1*ENTl1aCGMo16ob7Kc1lI2w.png

Understanding Data Types (cont.)

- **Effective Visualization**

- Choosing the right visualization technique depends on the data type.
 - **Numerical data** might be best represented through histograms, scatter plots, or line charts, allowing for the identification of patterns, trends, and outliers.
 - **Categorical data** is often displayed using bar charts or pie charts, which highlight the distribution and proportions of different categories.
- Mixing up visualization methods due to incorrect data type identification can lead to misleading or unintelligible results.



Understanding Data Types (cont.)

- **Correct methods for data cleaning and pre-processing**
 - Different data types require different cleaning and pre-processing techniques.
 - **Numerical data** may need outlier detection and handling, normalization, or transformation.
 - **Categorical data** might require encoding techniques to convert categories into a format that can be used in mathematical models.
 - **Text data** often needs to be tokenized, stemmed, or lemmatized for analysis.
 - Without accurately identifying data types, these pre-processing steps may be incorrectly applied, leading to poor analysis outcomes.



Source: <https://www.teraflow.ai/wp-content/uploads/2021/12/3-Big-Benefits-Of-Data-Cleansing--1024x675.jpg>

Understanding Data Types (cont.)

- **Effective feature engineering**
 - Understanding data types is crucial for effective feature engineering, which involves creating new variables from existing ones to improve model performance.
 - From a date-time variable (numerical or categorical), one might extract day of the week, month, or time of day as separate features.
 - Knowing the data type guides the feature engineering process, ensuring that meaningful features are created.



Understanding Data Types (cont.)

- **Suitable algorithms and models for better performance**
 - In machine learning, the choice of algorithm and the performance of the predictive model can be highly dependent on the nature of the input data.
 - Some algorithms work better with continuous numerical data, while others are more suited for categorical data.
 - Properly identifying the data types helps in selecting the most suitable model, leading to better performance and more accurate predictions.



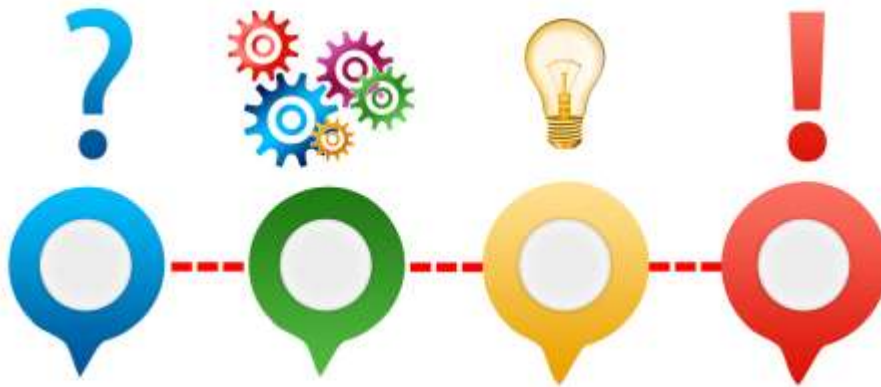
Understanding Data Types (cont.)

- **Ensuring data integrity**
 - Correctly identifying data types is essential for maintaining data integrity.
 - It ensures that operations performed on the data are valid and that the data conforms to the expected format.
 - Performing arithmetic operations on categorical data or treating numerical data as categorical without proper justification can lead to incorrect conclusions.



Outline

- Exploratory Data Analysis (EDA)
 - **EDA for Numerical Data**
 - EDA for Text Data
- Text Data Pre-processing



EDA for Numerical Data

- **Descriptive statistics** provide a powerful summary of data sets by giving insights into the **central tendency**, **dispersion**, and **distribution shape**.

- **Central tendency**

- **Mean:** the arithmetic average of a set of values

$$\mu = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

- For example,

- Consider the dataset: [4, 8, 6, 5, 3],

$$\mu = \frac{(4 + 8 + 6 + 5 + 3)}{5} = \frac{26}{5} = 5.2$$

EDA for Numerical Data (cont.)

- **Descriptive statistics** provide a powerful summary of data sets by giving insights into the **central tendency**, **dispersion**, and **distribution shape**.

- **Central tendency**

- **Mean:** the arithmetic average of a set of values
- **Median:** the middle value after all the values are put in an order list
- For example,
 - For the dataset: [3, 4, 5, 6, 8] (odd number of observations),

$$\text{median} = 5$$

- For the dataset: [3, 4, 5, 6] (even number of observations),

$$\text{median} = \frac{(4 + 5)}{2} = 4.5$$

EDA for Numerical Data (cont.)

- **Descriptive statistics** provide a powerful summary of data sets by giving insights into the **central tendency**, **dispersion**, and **distribution shape**.
 - **Central tendency**
 - **Mean**: the arithmetic average of a set of values
 - **Median**: the middle value after all the values are put in an order list
 - **Mode**: the most frequent occurring value in a set of values
 - For example,
 - In the dataset: [2, 3, 4, 4, 5, 5, 5, 6],

$$mode = 5$$

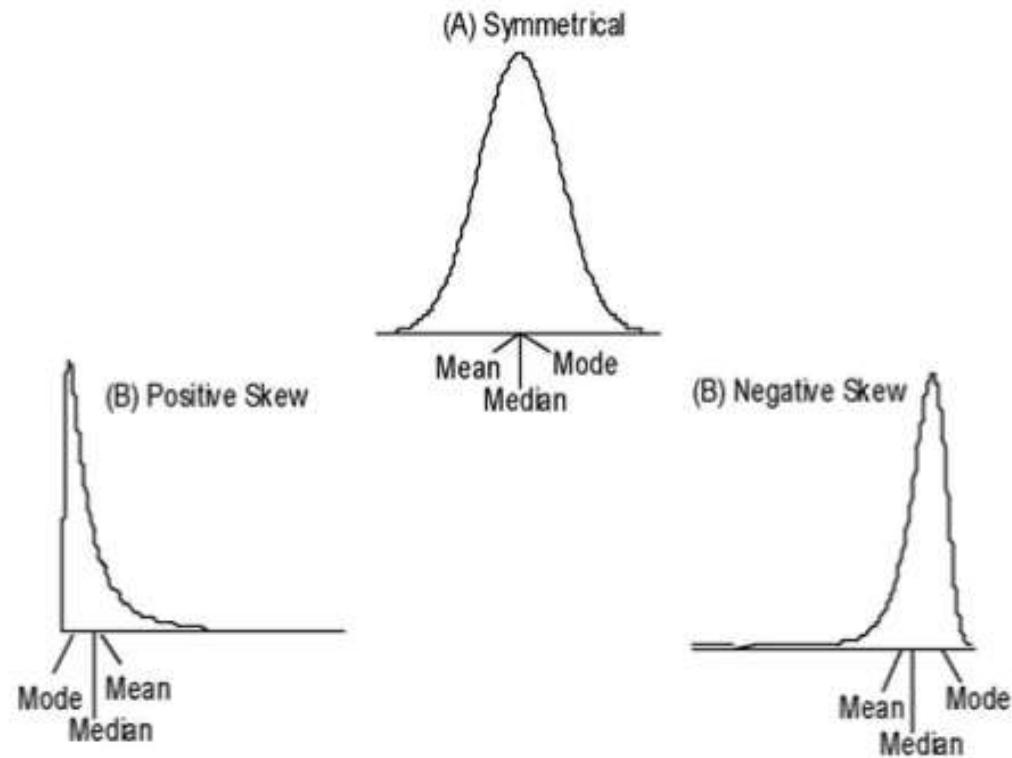
In-class participation

Question 1:

Among these three central tendency indicator, which one is the best to use when doing EDA for a new dataset?

EDA for Numerical Data (cont.)

- **Descriptive statistics** provide a powerful summary of data sets by giving insights into the [central tendency](#), [dispersion](#), and [distribution shape](#).
 - **Central tendency**



EDA for Numerical Data (cont.)

- **Descriptive statistics** provide a powerful summary of data sets by giving insights into the **central tendency**, **dispersion**, and **distribution shape**.

- **Range**

- The difference between the smallest and largest values in the data set.
- For example,
 - For the dataset: [4, 8, 6, 5, 3],

$$range = 8 - 3 = 5$$

- **Interquartile range**

Q0	the minimum
Q1	bigger than 25% of the data points
Q2	the median
Q3	bigger than 75% of the data points
Q4	the maximum

The inter-quartile range (IQR):

$$IQR = Q3 - Q1$$

EDA for Numerical Data (cont.)

- **Descriptive statistics** provide a powerful summary of data sets by giving insights into the **central tendency**, **dispersion**, and **distribution shape**.

- **Variance**

- Measures how much the data points differ from the mean.
- For example,
 - A simplified dataset: [4, 8, 6], first calculate the mean:

$$(4 + 8 + 6) / 3 = 6$$

Then, calculate each data point's squared difference from the mean:

$$[(4 - 6)^2, (8 - 6)^2, (6 - 6)^2] = [4, 4, 0]$$

The variance is the average of these squared differences:

$$(4 + 4 + 0) / 3 = 2.67$$

EDA for Numerical Data (cont.)

- **Descriptive statistics** provide a powerful summary of data sets by giving insights into the **central tendency**, **dispersion**, and **distribution shape**.
 - **Standard Deviation**
 - The square root of the variance, providing a measure of spread that is in the same units as the data.
 - It tells us how much data varies from the average.
 - Continuing from the variance example,

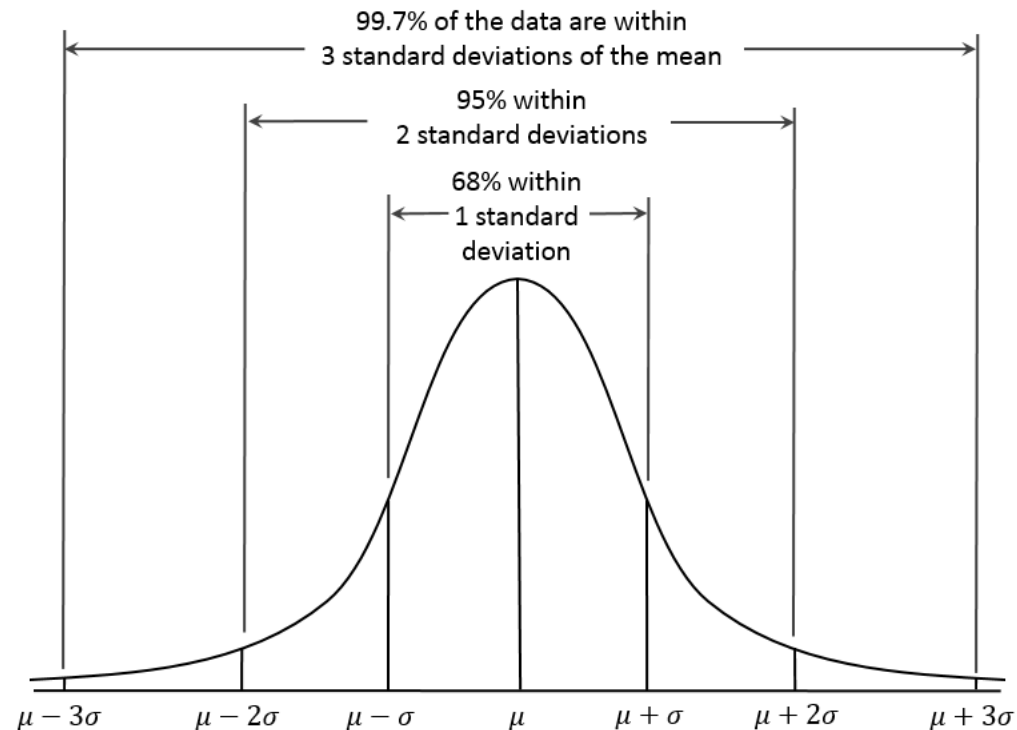
$$\text{Variance} = 2.67$$

$$\text{Standard Deviation} = \sqrt{2.67} \approx 1.63$$

EDA for Numerical Data (cont.)

- **Descriptive statistics** provide a powerful summary of data sets by giving insights into the **central tendency**, **dispersion**, and **distribution shape**.
 - **Standard Deviation**

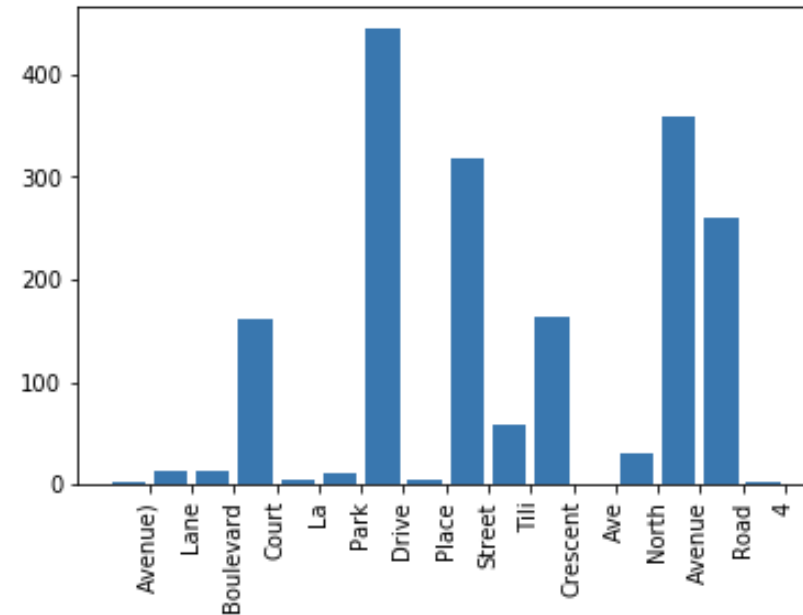
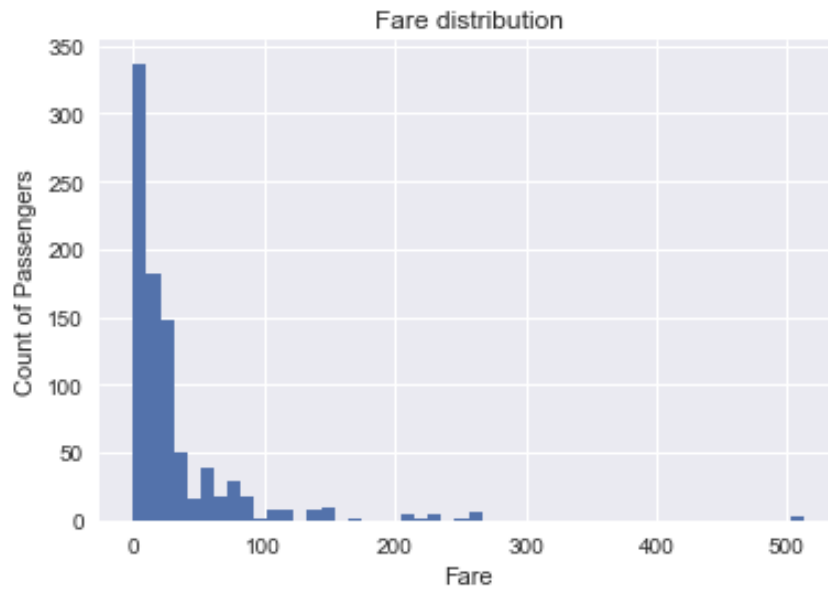
$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$



Visualization of Numerical Data

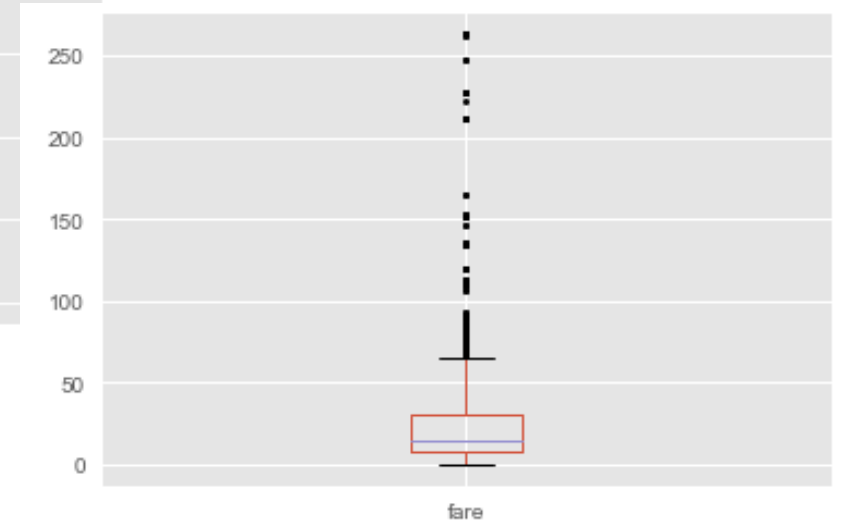
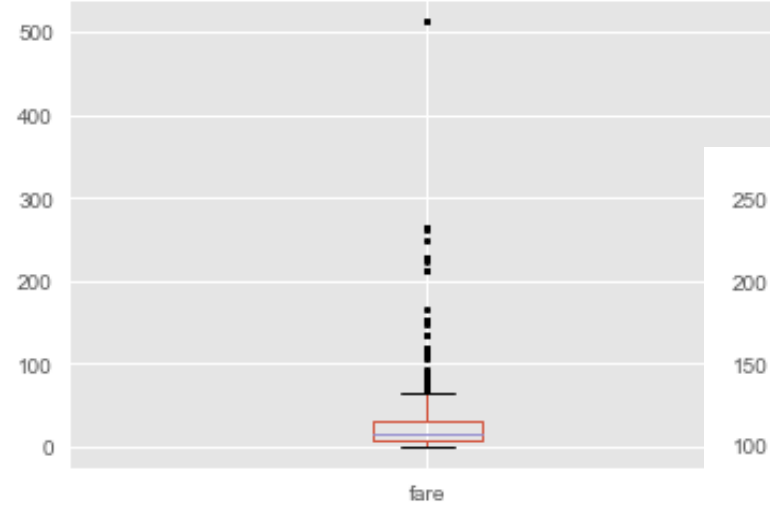
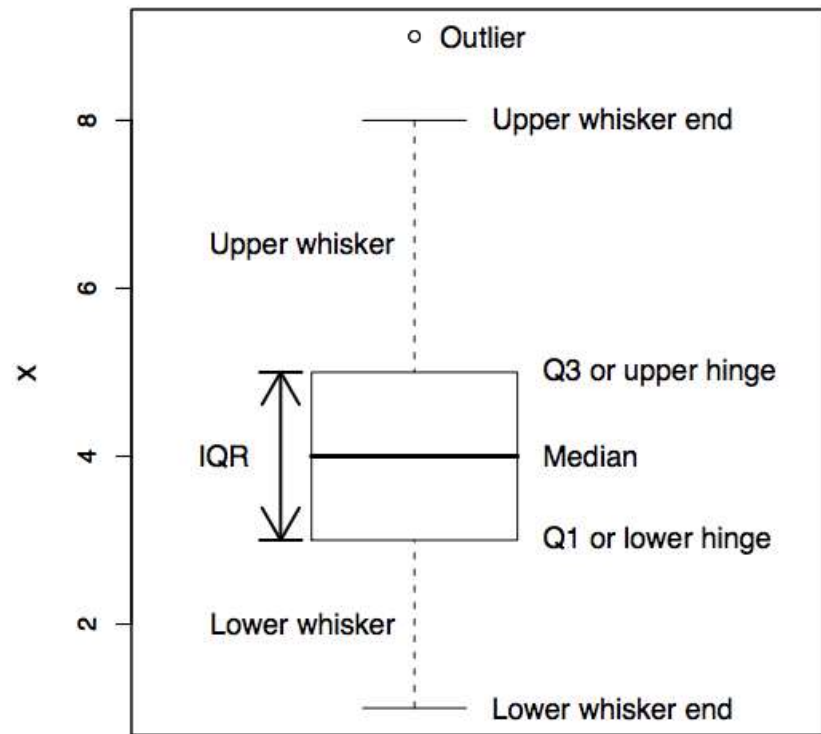
- **Histograms**

- a quick way of learning the characteristics of your data, including central tendency, spread, shape, outliers, etc



Visualization of Numerical Data (cont.)

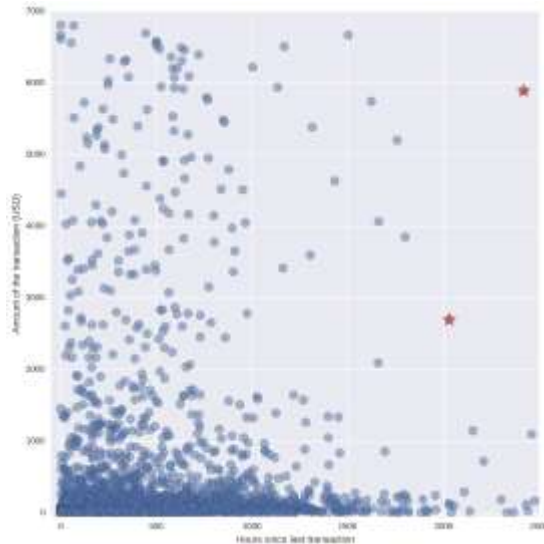
- **Boxplots (or Box-and-Whiskers Plot)**
 - display five-point summaries and **potential outliers** in graphical form.



Visualization of Numerical Data (cont.)

- Scatter plots

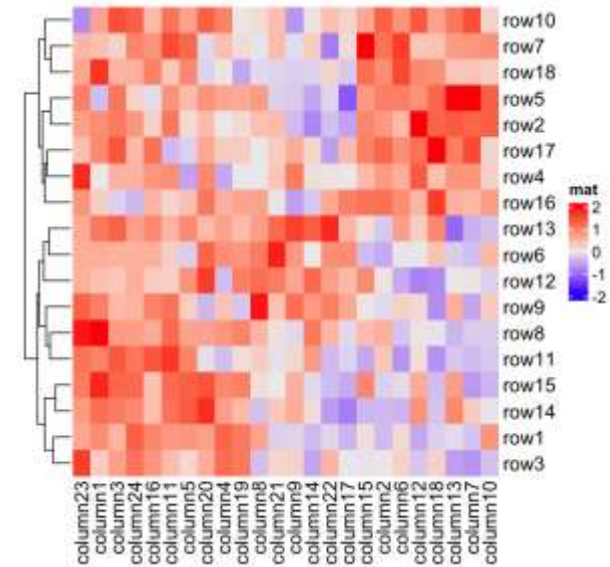
- a scatter matrix provides a matrix of scatter plots for each pair of attributes.



The Switzerland heart disease data set from [UCI machine learning data repository](https://archive.ics.uci.edu/ml/datasets/Heart+Disease)

- Heatmap

- to visualize the correlation matrix, showing relationships between variables using colour intensity



Source: https://jokergoo.github.io/ComplexHeatmap-reference/book/02-single_heatmap_files/figure-html/default-1.png

In-class participation

Question 2:

How to visualize data in high dimensionality?

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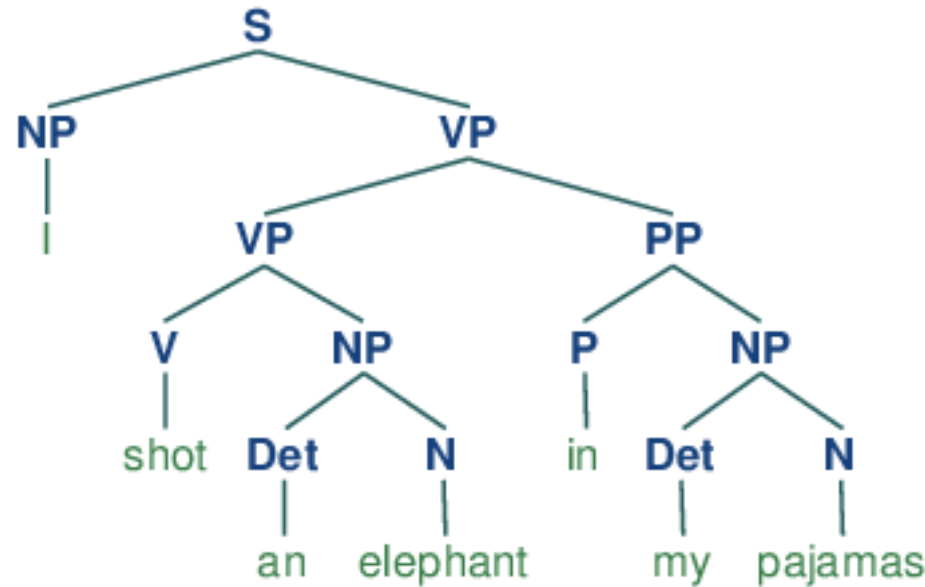


EDA for Character and Text Data

- Character and text data analysis, often referred to within the broader field of **Natural Language Processing (NLP)**, focuses on the processing and analysis of textual data to derive meaningful information.
- The **goal** of character and text data analysis is to provide understanding of how the text is processed without having a human read it.
- This analysis can range from simple tasks like counting word frequencies to more complex ones such as sentiment analysis, topic modelling, and language translation.
- The analysis of character and text data involves several steps and techniques, each designed to tackle the unique challenges presented by textual content.

EDA for Character and Text Data (cont.)

- **Syntax:** the structure of language, e.g., grammar rules
 - How individual words are composed to make well-formed sentences and paragraphs.
- Analyse sentence structure



S – sentence

N - noun

NP - noun phrase

V – verb

VP - verb phrase

P - preposition

PP - prepositional phrase

ADV - adverb

ADVP - adverb phrase

ADJ - adjective

ADJP - adjective phrase

Det - determiner

Source: The NLTK book

EDA for Character and Text Data (cont.)

- **Semantics:** the meaning of the individual words within the surrounding context
 - Understand the theme of a given text fragment.
 - Topical collocation

For sample,

Is "white house" a topical collocation?

- In a real-estimate context: compositional phrase
- In a political context: topical collocation

"Arts"	"Budgets"	"Children"	"Education"
NEW	MILLION	CHILDREN	SCHOOL
FILM	TAX	WOMEN	STUDENTS
SHOW	PROGRAM	PEOPLE	SCHOOLS
MUSIC	BUDGET	CHILD	EDUCATION
MOVIE	BILLION	YEARS	TEACHERS
PLAY	FEDERAL	FAMILIES	HIGH
MUSICAL	YEAR	WORK	PUBLIC
BEST	SPENDING	PARENTS	TEACHER
ACTOR	NEW	SAYS	BENNETT
FIRST	STATE	FAMILY	MANIGAT
YORK	PLAN	WELFARE	NAMPHY
OPERA	MONEY	MEN	STATE
THEATER	PROGRAMS	PERCENT	PRESIDENT
ACTRESS	GOVERNMENT	CARE	ELEMENTARY
LOVE	CONGRESS	LIFE	HAITI

The William Randolph Hearst Foundation will give \$1.25 million to Lincoln Center, Metropolitan Opera Co., New York Philharmonic and Juilliard School. "Our board felt that we had a real opportunity to make a mark on the future of the performing arts with these grants an act every bit as important as our traditional areas of support in health, medical research, education and the social services," Hearst Foundation President Randolph A. Hearst said Monday in announcing the grants. Lincoln Center's share will be \$200,000 for its new building, which will house young artists and provide new public facilities. The Metropolitan Opera Co. and New York Philharmonic will receive \$400,000 each. The Juilliard School, where music and the performing arts are taught, will get \$250,000. The Hearst Foundation, a leading supporter of the Lincoln Center Consolidated Corporate Fund, will make its usual annual \$100,000 donation, too.

Outline

- Exploratory Data Analysis (EDA)
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Data Pre-processing for Text Data

- Encoding techniques

- **One-hot encoding** is a method used to convert categorical data into a binary (0 or 1) matrix.

- Creating a binary column for each category of the variable and assigning a 1 or 0 in those columns for each data point, depending on which category is present.
- For example,

Imagine we have a dataset with a categorical feature "Colour" that includes three categories: Red, Blue, and Green.

Item	Color
1	Red
2	Blue
3	Green
4	Red

One-hot encoding



Item	Color_Red	Color_Blue	Color_Green
1	1	0	0
2	0	1	0
3	0	0	1
4	1	0	0

Data Pre-processing for Text Data (cont.)

- Encoding techniques

- **Label encoding** replaces the categorical values with a sequence of integers (e.g., 0, 1, 2, ...).
 - This method is particularly useful when the categorical variable is ordinal (i.e., the categories have a meaningful order or ranking).
 - For example,

Consider a dataset with a categorical feature "Size" that includes three categories: Small, Medium, and Large.

Item	Size
1	Small
2	Medium
3	Large
4	Small

Label encoding



Item	Size
1	0
2	1
3	2
4	0

Data Pre-processing for Text Data (cont.)

- **Vectorization methods**

- **Count vectorizer** is a method used in text processing that converts a collection of text documents into a matrix of token counts.
 - The primary idea is to represent text data numerically by counting the frequency of each word in the document, allowing algorithms to process natural language data.
 - How it works?
 1. **Tokenization:** The text is split into tokens (typically words or phrases).
 2. **Vocabulary Building:** A vocabulary of all unique tokens in the text corpus is created.
 3. **Count Calculation:** For each document, the frequency of each word in the vocabulary is counted and represented in a vector, where each dimension corresponds to a term in the vocabulary.

Example: Count Vectorizer

- Consider a simple dataset of two text documents:

1. "The cat sat on the mat."

2. "The dog sat on the log."

Step 1: Tokenise and normalise the text (convert to lowercase, remove punctuation).

- Document 1: ["the", "cat", "sat", "on", "the", "mat"]
- Document 2: ["the", "dog", "sat", "on", "the", "log"]

Step 2: Build a vocabulary of unique terms.

- Vocabulary: ["the", "cat", "sat", "on", "mat", "dog", "log"]

Step 3: Create a document-term matrix with the counts of each word appearing in each document.

	the	cat	sat	on	mat	dog	log
Doc 1	2	1	1	1	1	0	0
Doc 2	2	0	1	1	0	1	1

Tokenisation

- **Tokenization** is a fundamental step in natural language processing (NLP) and text analysis.
- A **text corpus** is broken down into smaller units, such as words, phrases, symbols, or other meaningful elements called **tokens**.
- The process helps in transforming text into a form that can be easily analysed or processed by algorithms.
- Tokenization is crucial for tasks like sentiment analysis, topic modelling, text classification, and many others, as it enables the conversion of unstructured text into a structured form.
- **Types of tokenisation**
 - Word Tokenization
 - Sentence Tokenization
 - Subword Tokenization
 - Character Tokenization

Tokenisation (cont.)

- **Importance of Tokenization**
 - **Pre-processing for NLP Tasks:** Tokenization is often the first step in preparing text for NLP tasks, allowing for further processing such as part-of-speech tagging, syntactic parsing, and entity recognition.
 - **Handling Large Texts:** By breaking down large texts into manageable units, tokenization makes it easier to analyse and process text data efficiently.
 - **Improving Model Performance:** Proper tokenization can significantly impact the performance of NLP models by ensuring that the input data is in a suitable format for analysis.
 - **Flexibility and Adaptability:** Different tokenization strategies can be adopted depending on the specific requirements of the task or the characteristics of the language being processed.

Data Pre-processing for Text Data (cont.)

- **Case Normalisation**

- Capitalisation helps readers differentiate
 - Common nouns: writer, teacher, cookies, ...
 - Proper noun: Herman Melville, Snoopy, University of Melbourne, ...
- Case normalisation converts all the words into either uppercase or lowercase words.
- *Case normalisation is not always needed!!!*
 - Information Retrieval: “Data Wrangler” v.s. "data wrangler"
 - Named Entity Recognition: One would better keep capitalised words left as capitalised.
 - A word may change its meaning
- One function to finish case normalisation: **lower()**

✓
0s

```
▶ text = "Case Normalisation for Text Data Pre-processing!"  
  text = text.lower()  
  print(text)
```

```
↳ case normalisation for text data pre-processing!
```

Data Pre-processing for Text Data (cont.)

- **Removing stop words**
 - **Stop words** are words that are extremely common and carry little lexical content.
 - They are often function words in English, for example,
 - articles (e.g., "a", "the", and "an")
 - pronouns (e.g., "he", "him", and "they")
 - particles (e.g., "well", "however" and "thus")
 - Stop words usually refer to the most common words in a language.
 - The general strategy for determining whether a word is a stop word or not is to compute its total number of appearances in a corpus.

Data Pre-processing for Text Data (cont.)

- **Removing stop words**
 - Why should we remove stop words?
 - Stop words usually appear to be of little value and have little impact on the final results, as the presence of stop words in a text document does not really help distinguishing it from other documents.
 - Failing to remove those common words could lead to skewed analysis results.
 - For example, Email analysis: remove headers (e.g., "Subject", "To", and "From"), remove a lengthy legal disclaimer

Data Pre-processing for Text Data (cont.)

- **Stemming**

- The process of identification and removal of prefixes, suffixes, and pluralisation, which leaves you with a stem.
 - 'watches -> watch'
 - 'parties -> party'
 - 'carrying -> carry'

Data Pre-processing for Text Data (cont.)

- **Lemmatization**

- A more advanced form of stemming that makes use of, for example, the context surrounding the words, an existing vocabulary, morphological analysis of words and other grammatical information (e.g., part-of-speech tags) to determine the basic or dictionary form of a word, which is known as the lemma.

```
▶ from nltk.stem import WordNetLemmatizer

# Initialize the WordNetLemmatizer
lemmatizer = WordNetLemmatizer()

# Lemmatize some words
print("Lemmatized Dog:", lemmatizer.lemmatize("dogs"))
print("Lemmatized Running (as a verb):", lemmatizer.lemmatize("running", pos='v')) # 'v' denotes verb
```

⇒ Lemmatized Dog: dog
Lemmatized Running (as a verb): run

Summary & To-do List

- Please download and read the materials provided on Moodle.
- Review the content learnt from Week 4.
- Assessments
 - Form a group for Assessment 1
 - Continue Assessment 1 and record your collaboration, if any.
- Next week: Data discovery and collection